Imperial College London: Department of Aeronautics
AERO-60001 Aerodynamics 3

Tutorial Sheet 2 (Compressible Flow)

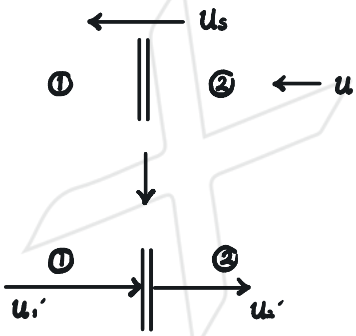
1. A plane normal shock wave travels at speed us through stationary air where the pressure and temperature are p_1 and T_1, respectively. Show that the velocity u induced behind the shock is given by,

$$u = \frac{2a_1}{\gamma+1}\left[M_s - \frac{1}{M_s}\right]$$

where $M_s = u_s/a_1$.

A Eurostar train from London to Paris enters the tunnel at a speed of $144\,\mathrm{km\,hr^{-1}}$. Assuming that there is no leakage of air past the train, calculate the Mach number of the shock-wave that forms in front of the train and the pressure ratio across it. Assume atmospheric temperature of 300 K.

[Ans: M = 1.071, Pressure ratio = 1.17]



$$\therefore\ u = u_s - u_2'$$

$$M_s = \frac{u_s - u_1}{a_1} = \frac{u_s}{a_1} = M_1'$$

$$\because\ \frac{\rho_2}{\rho_1} = \frac{u_1'}{u_2'} = \frac{(\gamma+1)(M_1')^2}{(\gamma-1)(M_1')^2+2}$$

$$\therefore\ u = u_s - u_1'\frac{u_2'}{u_1'} = u_s - u_s\frac{(\gamma-1)M_s^2+2}{(\gamma+1)M_s^2}$$

$$= M_s a_1 \frac{(\gamma+1)M_s^2 - (\gamma-1)M_s^2 - 2}{(\gamma+1)M_s^2}$$

$$= M_s a_1 \frac{2M_s^2 - 2}{(\gamma+1)M_s^2}$$

$$= \frac{2a_1}{\gamma+1}\left(M_s - \frac{1}{M_s}\right)$$

for $u = 144\ \mathrm{km/h} = 40\ \mathrm{m/s}$ and $a_1 = \sqrt{\gamma R T_1} = 14\sqrt{615}\ \mathrm{m/s}$

$$\therefore\ 40 = \frac{2\gamma\sqrt{615}}{2.8}\left(M_s - \frac{1}{M_s}\right)$$

$$\therefore\ 0.16130 = M_s - \frac{1}{M_s}$$

$$\therefore\ M_s^2 - 0.16130 M_s - 1 = 0$$

$$\because\ M_s > 0$$

$$\therefore\ M_s = \frac{0.16130 + \sqrt{(-0.16130)^2 + 4}}{2} = 1.084$$

$$\therefore\ \frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1}\left(M_s^2 - 1\right)$$

$$= 1.175$$

2. A piston accelerates from rest along a very long tube. Initially the air in the tube is stationary. The pressure and speed of sound are p_0 and a_0, respectively. Show that, provided no reflections occur from the far end of the tube, the speed of sound on the back face of the piston is given by,

$$a = a_0 - \frac{\gamma - 1}{2} u_p$$

where u_p is the instantaneous speed of the piston.

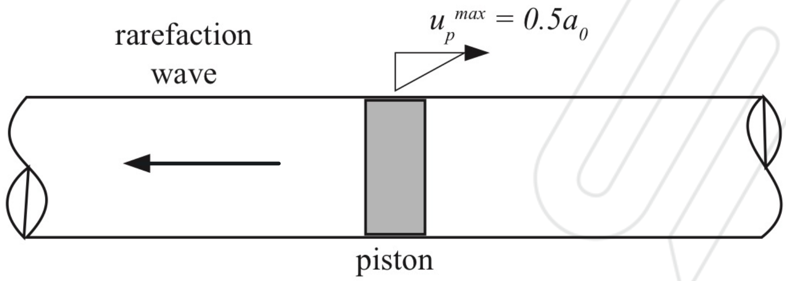
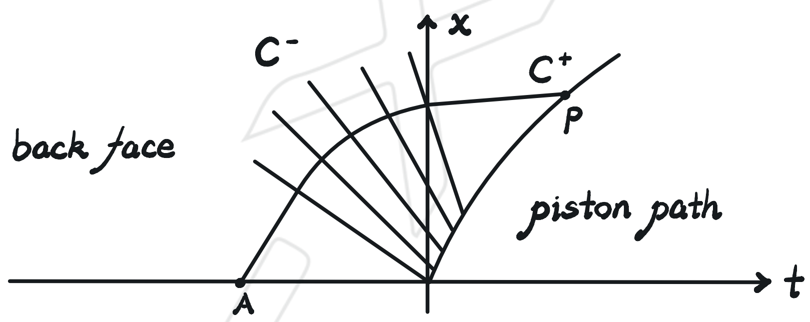


Figure 1

The piston accelerates to a maximum velocity is $up^{max} = 0.5a_0$. Calculate the corresponding pressure and local Mach number of the flow.

[Ans: $p/p_0 = 0.478$, $M = 0.556$]



For unsteady. isentrophic flow along C^+ as shown in graph

$$\Gamma^+ = u_0 + \frac{2a_0}{\gamma - 1} = u_p + \frac{2a_p}{\gamma - 1} = \text{constant}$$

$$\therefore 2a_0 = (\gamma - 1) u_p + 2a_p$$

$$\therefore a_p = a_0 - \frac{\gamma - 1}{2} u_p$$

$$\therefore a = a_0 - \frac{\gamma - 1}{2} u_p$$

At u_p^{max}, $u_p^{max} = 0.5a_0$

$$\therefore a = (1 - \frac{\gamma - 1}{4}) a_0 = \frac{5 - \gamma}{4} a_0 = 0.9 a_0$$

$$\therefore \frac{p_0}{p} = (\frac{T_0}{T})^{\frac{\gamma}{\gamma - 1}} = (\frac{a_0}{a})^{\frac{2\gamma}{\gamma - 1}} = 2.0908$$

$$\therefore \frac{p}{p_0} = 0.4783$$

$$\therefore M = \frac{u_p}{a} = \frac{0.5a_0}{a_0 \frac{a}{a_0}}$$

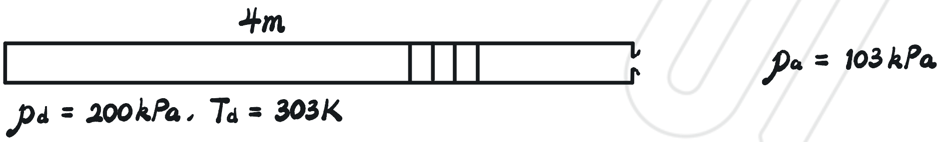
$$= 0.5 \frac{a_0}{a}$$

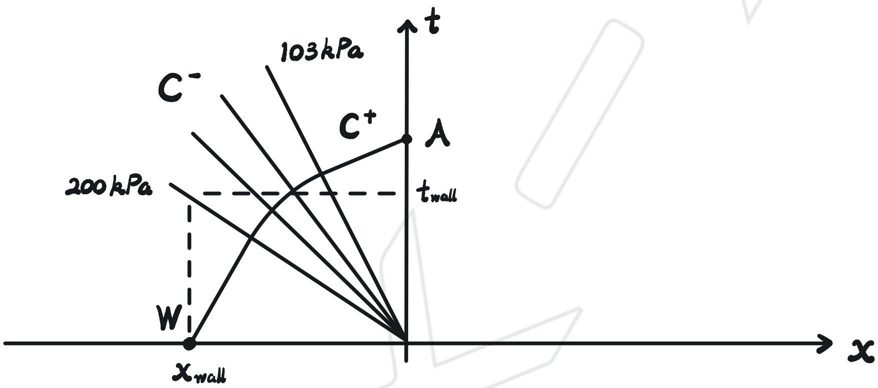
$$= 0.5 \frac{1}{0.9}$$

$$= \frac{5}{9} = 0.556$$

3. A diaphragm at the end of 4 m long pipe containing air at a pressure of 200 kPa and a temperature of 303 K suddenly ruptures causing an expansion wave to propagate down the pipe. Find the velocity at which air is discharged from the pipe if the ambient pressure is 103 kPa. Also find the speed of sound at the front and back of the wave and hence the time taken by the front of the wave to reach the end of the pipe.

[Ans: V = 158 ms^{-1}, Speed of sound at the front = 348 ms^{-1}, at the back = 317.5 ms^{-1}, time taken to reach the end is 0.0115 s]





$$a_d = \sqrt{\gamma R T_d} = 348.920 \text{ (m/s)} = a_W$$

$$a_a = \frac{a_a}{a_d} a_d = \left(\frac{p_a}{p_d}\right)^{\frac{\gamma-1}{2\gamma}} a_d = 317.363 \text{ (m/s)} = a_A$$

Along C^+,

$$\Gamma^+ = u_W + \frac{2a_W}{\gamma-1} = u_A + \frac{2a_A}{\gamma-1}$$

$$\therefore 0 + 5a_W = u_A + 5a_A$$

$$\therefore u_A = 5a_W - 5a_A = 157.785 \text{ (m/s)}$$

Along C^-,

$$\frac{dt}{dx} = \frac{1}{u_d - a_d} = -2.86599 \times 10^{-3} \text{ (s/m)}$$

$$\therefore t = \int \frac{dt}{dx} dx = -2.86599 \times 10^{-3} x \quad \text{since } x = 0 \text{ when } t = 0$$

$$\therefore t_{wall} = -2.86599 \times 10^{-3} \cdot (-4) = 0.011464 \text{ (s)}$$

4. Consider a long tube of constant cross-sectional area containing a driver gas and a driven gas separated by a diaphragm as shown in the figure. At time $t = 0$, the flow variables for the driver gas are p_4, T_4, $u_4 = 0$. The driven gas has properties p_1, T_1 and $u_1 = 0$. Note that $p_4 > p_1$, T_4 is not equal to T_1 and that the ratio of specific heats of the gases are not the same. $\gamma_4 \neq \gamma_1$

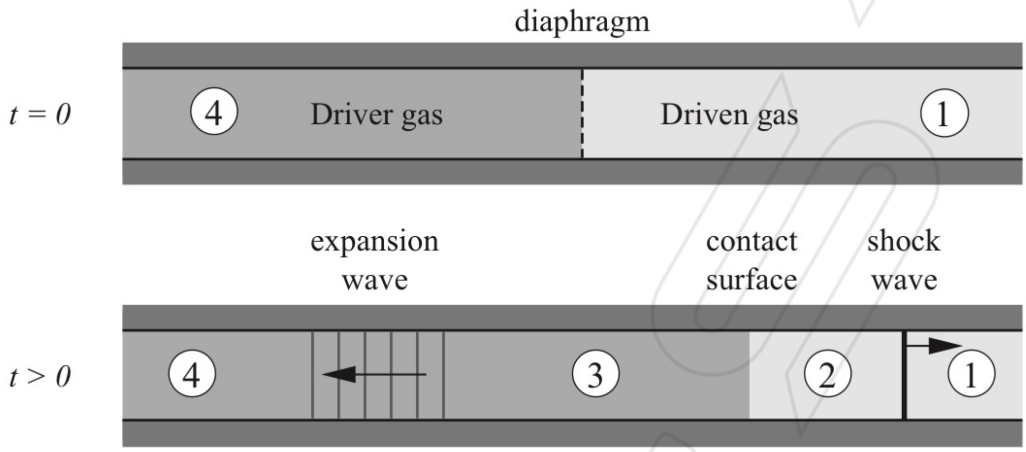
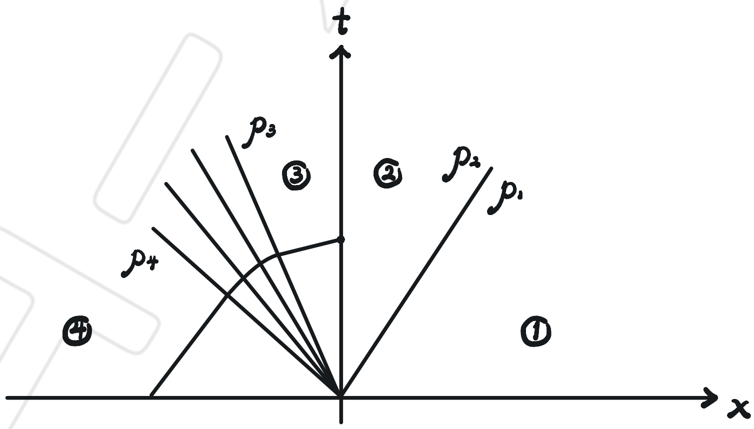


Figure 2

The diaphragm is suddenly ruptured and the flow develops as described in lectures. Derive a relation for the pressure ratio p_4/p_1 in terms of the shock Mach number (M_s), the speed of sound in regions 1 and 4 and the ratio of specific heats of the two gases. You may assume that the gases do not mix and a slip line (also known as contact discontinuity) exists across which the pressure and velocity are the same but the entropy, density and temperature are not the same.

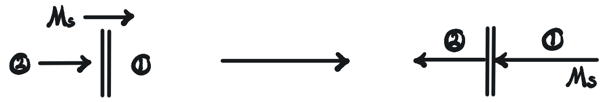
[Ans: ...]



$$\frac{p_4}{p_1} = f(M_s, a_1, a_4, \gamma_1, \gamma_4)$$

Acrossing the shock wave,

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma_1}{\gamma_1 + 1}(M_s^2 - 1)$$



$$\therefore \frac{\rho_2}{\rho_1} = \frac{u_1'}{u_2'} = \frac{(\gamma_1+1)M_s^2}{(\gamma_1-1)M_s^2+2}$$

$$\therefore u_2 = u_s - u_2' = u_s - u_1'\frac{u_2'}{u_1'}$$

$$= u_s - u_s\frac{(\gamma_1-1)M_s^2+2}{(\gamma_1+1)M_s^2}$$

$$= u_s\frac{2M_s^2-2}{(\gamma_1+1)M_s^2}$$

$$= \frac{2a_1}{\gamma_1+1}\left(M_s - \frac{1}{M_s}\right)$$

Acrossing the slip line,

$p_3 = p_2$, $u_3 = u_2$

Acrossing expansion wave, for C^+

$$\Gamma^+ = u_4 + \frac{2a_4}{\gamma_4-1} = u_3 + \frac{2a_3}{\gamma_4-1} = \text{constant}$$

$$\therefore a_3 = a_4 - \frac{\gamma_4-1}{2}u_3 = a_4 - \frac{\gamma_4-1}{\gamma_1-1}a_1\left(M_s - \frac{1}{M_s}\right)$$

$$\therefore \frac{p_4}{p_3} = \left(\frac{a_4}{a_3}\right)^{\frac{2\gamma_4}{\gamma_4-1}}$$

$$\therefore \frac{p_4}{p_1} = \frac{p_4}{p_3}\frac{p_2}{p_1}$$

$$= \frac{1+\frac{2\gamma_1}{\gamma_1+1}(M_s^2-1)}{\left[1-\frac{a_1}{a_4}\frac{\gamma_4-1}{\gamma_1-1}\left(M_s-\frac{1}{M_s}\right)\right]^{\frac{2\gamma_4}{\gamma_4-1}}}$$